

- 6 Horseradish peroxidase (HRP) is an enzyme that can be extracted from the roots of the horseradish plant, *Armoracia rusticana*. HRP is used extensively in industry and technology.

In the reaction catalysed by HRP, hydrogen peroxide (H_2O_2) is used to oxidise an organic substrate. This is summarised in Fig. 6.1.

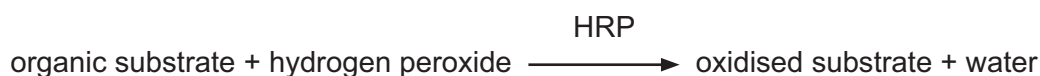


Fig. 6.1

- (a) Inhibitors can have an effect on V_{max} , the maximum rate of reaction, and K_m , the Michaelis–Menten constant, of HRP.

Fig. 6.2 shows the effect of substrate concentration on the rate of reaction of HRP.

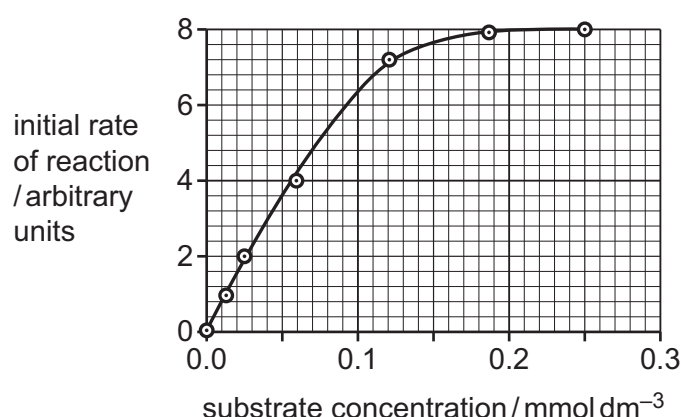


Fig. 6.2

Complete Fig. 6.2 by drawing a curve to show how the presence of a non-competitive inhibitor will affect the rate of reaction of HRP.

Use the curve **you have drawn** to obtain an estimate of K_m .

$K_m = \dots\dots\dots$ [2]

- (b) Scientists can design synthetic DNA nucleotide sequences to produce a synthetic HRP gene. These sequences will include a start codon and a stop codon so that translation of messenger RNA (mRNA) can occur.

Explain what is meant by a start codon **and** a stop codon.

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..... [2]

- (c) HRP is used in an immunological test known as a sandwich ELISA. One use of the test is to diagnose disease.

Fig. 6.3 outlines the main steps in a sandwich ELISA in which a toxin released by a pathogen is detected in a sample of body fluid taken from a person who is ill.

The test involves two types of monoclonal antibody that can bind to the toxin, a capture antibody and a detection antibody.

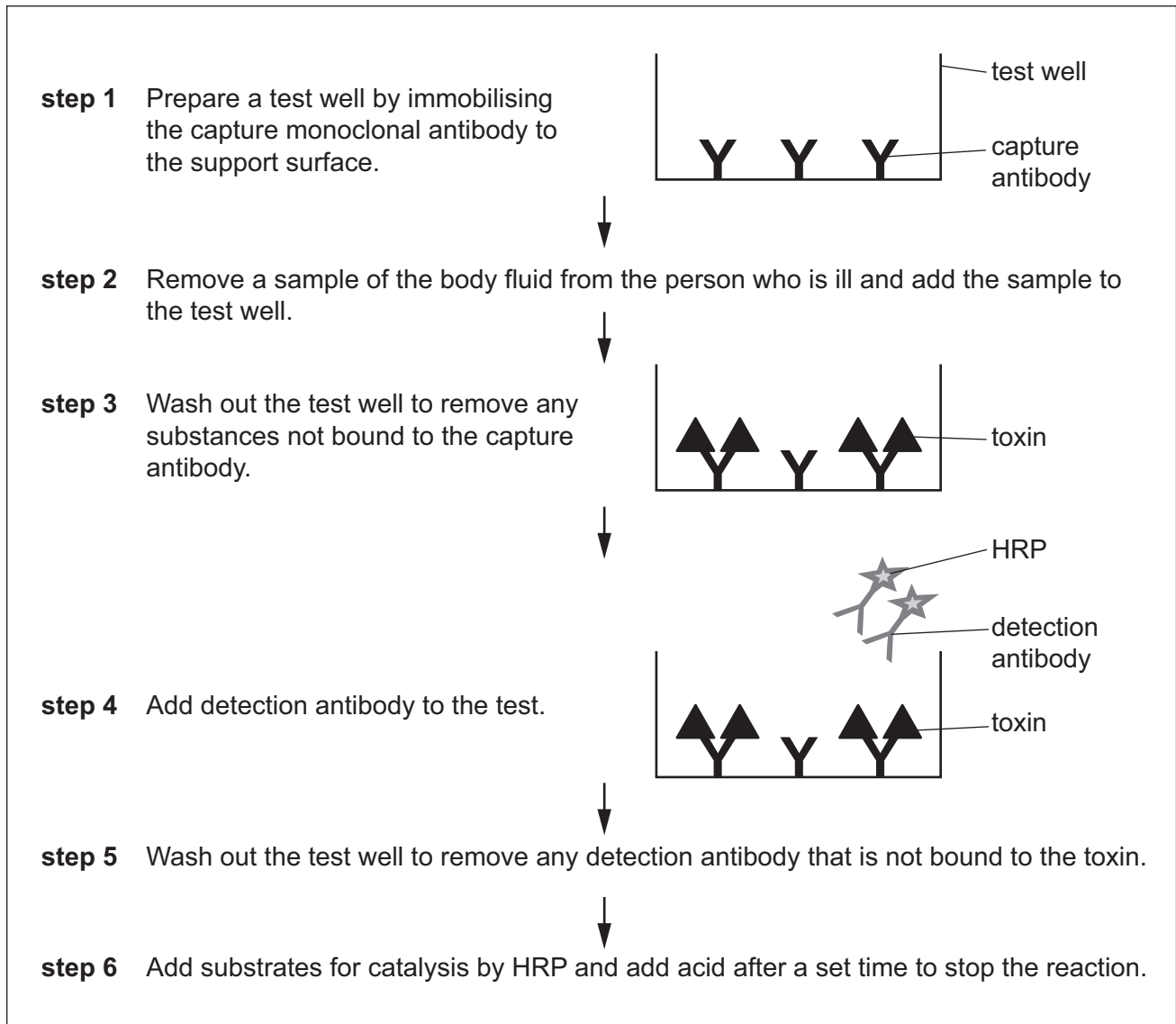


Fig. 6.3

- (i) Complete Fig. 6.4 to show the situation in the test well after **step 5**.



Fig. 6.4

- (ii) With reference to Fig. 6.3, describe **one** difference between the capture antibody and the detection antibody.

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- (iii) Fig. 6.5 outlines the reaction catalysed by HRP in **step 6**. The organic substrate, TMB, is in excess and changes colour when it is oxidised. This indicates a positive test result.

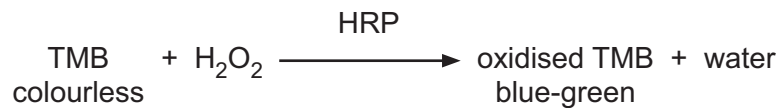


Fig. 6.5

A student suggested that:

- a low concentration of toxin may be diagnosed as a negative result instead of a positive result
- using a colorimeter after **step 6** in the sandwich ELISA test would provide a quantitative result and help to avoid this error.

With reference to Fig. 6.5, explain why using a colorimeter would provide a quantitative measurement for detection of a low concentration of the toxin and help to confirm a positive result.

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[Total: 10]